

CS 506: Introduction to Quantum Computing

Geometric Interpretation of a Qubit, Unitary Time Evolution, the Schrödinger Equation, and the Composite-System and Measurement Postulates

Scribe: Rownak Islam

September 9, 2026

1 Overview

First half: geometric (Bloch sphere) picture of a qubit — two angles instead of two amplitudes. Second half: the dynamics. Closed systems evolve under unitary operators; this comes straight out of the Schrödinger equation, and we check that the resulting evolution operator is actually unitary. Then two more postulates: combining qubits (tensor product, not concatenation), and measurement with respect to a basis, including the fact that measuring destroys the state.

2 Geometric Interpretation of a Qubit

2.1 Motivation

So far a qubit state is just a normalized vector of amplitudes, $\alpha_0 |0\rangle + \alpha_1 |1\rangle$. Fine algebraically, but without a geometric picture it's hard to see what a "rotation" of a qubit even means. Goal: rewrite a general qubit state using two angles, so every qubit state becomes a point on a sphere, and rotations become literal rotations.

2.2 Real Amplitudes and the Half-Angle

General qubit:

$$\alpha_0 |0\rangle + \alpha_1 |1\rangle, \quad |\alpha_0|^2 + |\alpha_1|^2 = 1.$$

Restricting to real amplitudes for now, a natural choice satisfying the normalization constraint is

$$\alpha_0 = \cos\left(\frac{\theta}{2}\right), \quad \alpha_1 = \sin\left(\frac{\theta}{2}\right),$$

since $\cos^2(\theta/2) + \sin^2(\theta/2) = 1$ automatically, for any θ . We use $\theta/2$ and not just θ because in the geometric (rotation) picture, a change of $\theta/2$ in the state ends up corresponding to an actual rotation of θ on the sphere — a standard quirk of how qubits relate to rotations. So at this point we have

$$\cos\left(\frac{\theta}{2}\right) |0\rangle + \sin\left(\frac{\theta}{2}\right) |1\rangle.$$

2.3 Allowing Complex Amplitudes

The above only covers real amplitudes, but qubit amplitudes are generally complex. To allow that, put an independent phase on each term:

$$e^{i\theta_1} \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\theta_2} \sin\left(\frac{\theta}{2}\right) |1\rangle.$$

Euler's identity: $e^{ix} = \cos x + i \sin x$, so $|e^{ix}|^2 = \cos^2 x + \sin^2 x = 1$ — a phase factor always has modulus 1. More generally, for any complex β ,

$$|\beta e^{ix}|^2 = |\beta|^2 |e^{ix}|^2 = |\beta|^2.$$

So multiplying an amplitude by a unit-modulus phase never changes the probability attached to it.

2.4 Global Phase vs. Local Phase

Factor $e^{i\theta_1}$ out of both terms:

$$e^{i\theta_1} \left(\cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i(\theta_2 - \theta_1)} \sin\left(\frac{\theta}{2}\right) |1\rangle \right).$$

Let $\varphi = \theta_2 - \theta_1$, so this becomes

$$e^{i\theta_1} \left(\cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle \right).$$

Here $e^{i\theta_1}$ is the **global phase** (multiplies the whole state) and $e^{i\varphi}$ is the **local phase** (only multiplies the $|1\rangle$ term). The global phase can always be dropped — it has no observable effect, since it can be absorbed by rotating the whole vector, and doesn't change any measurement outcome. The local phase is different: since it applies to only one term, it changes the relative phase between $|0\rangle$ and $|1\rangle$, which does matter physically.

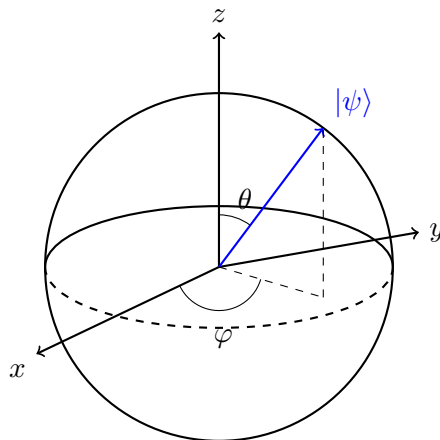
Dropping the global phase, the general qubit state is

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |1\rangle.$$

This is the geometric interpretation we were after: every physically distinct qubit state is just two real angles, θ and φ .

2.5 The Bloch Sphere

(θ, φ) are just spherical coordinates, so every qubit state can be plotted as a point on the unit sphere — the **Bloch sphere**. θ is the polar angle, measured from the z -axis. φ is the azimuthal angle, measured in the xy -plane from the x -axis to the projection of $|\psi\rangle$ onto that plane.



Changing θ and φ sweeps out every point on the sphere — every possible qubit state (up to the global phase already dropped) shows up somewhere on it.

3 Time Evolution of a Closed Quantum System

3.1 Unitary Evolution

Postulate: the Hilbert space of a system is made up of its observables, and unlike in Heisenberg's original picture, time doesn't act as an operator on the state — just a parameter. A closed quantum system evolves from an old state to a new state as

$$|\varphi\rangle_{\text{new}} = U |\psi\rangle_{\text{old}},$$

where U is a **unitary operator**: $\|U\| = 1$ and $UU^\dagger = I$.

3.2 Unitary Evolution Preserves the Norm

Probabilities need to sum to 1 both before and after evolution, so unitarity had better preserve the norm — check:

$$\begin{aligned} \|U |\psi\rangle\|^2 &= (U |\psi\rangle)^\dagger (U |\psi\rangle) \\ &= \langle\psi| U^\dagger U |\psi\rangle \\ &= \langle\psi| I |\psi\rangle \\ &= \langle\psi|\psi\rangle = \|\psi\|^2 = 1. \end{aligned}$$

So $\|\varphi\|^2 = \|U |\psi\rangle\|^2 = 1$ — the evolved state comes out normalized too, exactly as it should.

3.3 Linearity

Quantum mechanics is linear, which is what makes this algebra manageable: once an operator's action on the basis states is known, its action on any superposition just follows by distributing over the sum. For a qubit $\alpha_0 |0\rangle + \alpha_1 |1\rangle$ and a unitary gate U ,

$$U(\alpha_0 |0\rangle + \alpha_1 |1\rangle) = \alpha_0 U|0\rangle + \alpha_1 U|1\rangle.$$

4 The Schrödinger Equation

4.1 Statement

The Schrödinger equation (non-relativistic version) describes how a quantum state $|\psi(t)\rangle$ changes with time t :

$$i\hbar \frac{d|\psi(t)\rangle}{dt} = H |\psi(t)\rangle.$$

H is the Hamiltonian — the operator for the system's total energy (potential plus kinetic). H is also Hermitian, $H^\dagger = H$: that's a separate property from what it's called, and it holds because H 's eigenvalues are the system's energy levels, which are measurable and therefore have to be real. We use this Hermitian property below.

4.2 From Schrödinger to a Unitary Evolution Operator

Solving the Schrödinger equation over a short time step gives back exactly the unitary evolution $|\varphi\rangle = U |\psi\rangle$ from Section 3 — the reason the Schrödinger picture and the discrete unitary-operator picture are just two ways of describing the same physics.

Take two times $t_1 \approx t_2$, with $\delta = t_2 - t_1 \approx 0$, so over that short interval the Hamiltonian barely changes: $H(t_1) \approx H(t_2) = H$. Plugging into the Schrödinger equation,

$$i\hbar \frac{d|\psi(t_1)\rangle}{dt} = H |\psi(t_2)\rangle \implies \frac{d|\psi(t)\rangle}{dt} = \frac{H}{i\hbar} |\psi(t)\rangle = -i \frac{H}{\hbar} |\psi(t)\rangle.$$

This is exactly the ODE $\frac{dy}{dt} = \lambda y$, whose solution is $y = e^{\lambda t}$; here $\lambda = -iH/\hbar$. So

$$\boxed{|\psi(t_2)\rangle = e^{-i \frac{H}{\hbar} (t_2 - t_1)} |\psi(t_1)\rangle.}$$

Reading $|\psi(t_1)\rangle$ as the old state and $|\psi(t_2)\rangle$ as the new state, this is exactly $|\varphi\rangle_{\text{new}} = U |\psi\rangle_{\text{old}}$ with

$$U = e^{-i \frac{H}{\hbar} (t_2 - t_1)}.$$

4.3 Proof that U Is Unitary

Write $\delta' = (t_2 - t_1)/\hbar$ — a constant — so $U = e^{-i\delta' H}$. Since H is Hermitian,

$$U^\dagger = \left(e^{-i\delta' H}\right)^\dagger = e^{i\delta' H^\dagger} = e^{i\delta' H}.$$

U and $U^\dagger$ are both functions of the same operator H , so they commute and the exponents just add:

$$UU^\dagger = e^{-i\delta' H} e^{i\delta' H} = e^0.$$

From the matrix exponential series $e^X = I + X + \frac{X^2}{2!} + \dots$, every term past the identity vanishes at $X = 0$, so $e^0 = I$. Hence

$$UU^\dagger = I,$$

which is exactly unitarity. So the evolution operator coming out of the Schrödinger equation is guaranteed unitary.

5 Composite Systems Postulate

5.1 Tensor Product, Not Concatenation

Given two qubits, it's tempting to combine their states by just gluing the labels together — treating $|0\rangle + |1\rangle$ as $|01\rangle$ — but that's wrong. The correct way to combine independent quantum systems is the **tensor product**:

$$|0\rangle \otimes |1\rangle.$$

Combining a 1-qubit system with another 1-qubit system gives one 2-qubit (4-dimensional) composite system — dimensions multiply, not add.

5.2 Sanity Check

For 2 qubits, the computational basis is

$$|00\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |01\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |10\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |11\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Checking the tensor product against this:

$$|0\rangle \otimes |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle.$$

Matches the basis vector $|01\rangle$ above, so the tensor product does the right thing.

More generally, $|\psi_1\rangle \otimes |\psi_2\rangle$ is often just written $|\psi_1, \psi_2\rangle$ — combining independent systems this way is basically like multiplying independent probabilities.

6 Measurement (Observation) Postulate

6.1 Measurement Is With Respect to a Basis

Measurement is always with respect to some chosen basis — not automatic. As an example, take

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle,$$

and measure in the computational basis $\{|0\rangle, |1\rangle\}$. The outcome probabilities are just the amplitudes squared:

$$\Pr[|0\rangle] = \left(\frac{1}{\sqrt{3}}\right)^2 = \frac{1}{3}, \quad \Pr[|1\rangle] = \left(\sqrt{\frac{2}{3}}\right)^2 = \frac{2}{3}.$$

6.2 Measurement Destroys the Quantum State

Important fact: measurement is destructive. Once $|\psi\rangle$ is measured and some outcome comes up, the state collapses to exactly that basis vector — the original superposition is gone for good, not recoverable by measuring again.

6.3 Example: Measuring in the Hadamard Basis

Now measure the *same* $|\psi\rangle$, but in the Hadamard basis instead:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \quad |-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle.$$

Adding and subtracting these two gives $|+\rangle + |-\rangle = \sqrt{2}|0\rangle$ and $|+\rangle - |-\rangle = \sqrt{2}|1\rangle$, so inverting to write the computational basis in terms of the Hadamard basis:

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, \quad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}.$$

Substituting into $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$:

$$|\psi\rangle = \frac{1}{\sqrt{3}} \cdot \frac{|+\rangle + |-\rangle}{\sqrt{2}} + \sqrt{\frac{2}{3}} \cdot \frac{|+\rangle - |-\rangle}{\sqrt{2}} = \frac{1}{\sqrt{6}}(|+\rangle + |-\rangle) + \frac{1}{\sqrt{3}}(|+\rangle - |-\rangle),$$

using $\sqrt{2/3}/\sqrt{2} = 1/\sqrt{3}$. Collecting terms,

$$|\psi\rangle = \left(\frac{1}{\sqrt{6}} + \frac{1}{\sqrt{3}}\right)|+\rangle + \left(\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{3}}\right)|-\rangle.$$

So measuring $|\psi\rangle$ in the Hadamard basis collapses it to $|+\rangle$ or $|-\rangle$, with

$$\Pr[|+\rangle] = \left(\frac{1}{\sqrt{6}} + \frac{1}{\sqrt{3}}\right)^2, \quad \Pr[|-\rangle] = \left(\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{3}}\right)^2.$$

These still add up to 1, as they have to — same physical state, just different probabilities depending on which basis it's measured in.